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Fluid control valve unit and valve timing change device

Abstract

A fluid control valve unit includes: a fluid control valve V, including: an inlet **74**; a communication port **75, 76**; a sleeve **70** having a bottom end **70***a* and an opening end **70***b* and defining an axis S; and a spool **80** disposed in the sleeve to open and close the communication port; a cylindrical passage member **50**, having: an inner circumferential surface **51**, fit with the sleeve; a passage; an annular receiving part **53**; and an opening part **52**; a filter **60**; and a stopper **110**, receiving the opening end **70***b* of the sleeve **70** accommodated in the passage member **50**. The filter **60** is integrally formed by a metal plate spring member to include an annular plate part **61**, a filter part **62** surrounded by the annular plate part, and a plate spring piece extending from the annular plate part.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) This application claims the priority benefit of Japan Application No. 2023-087049, filed on May 26, 2023. The entirety of the above-mentioned patent application is hereby incorporated by reference herein and made a part of this specification.

BACKGROUND

Technical Field

(2) The invention relates to a fluid control valve unit and a valve timing change device of an internal combustion engine using the fluid control valve unit.

Description of Related Art

(3) As a conventional fluid control valve unit, a hydraulic pressure control valve including the following is known: a valve body, having an insertion hole with an axis as the center; a bottomed

cylindrical sleeve, closely inserted into the valve body; a spool valve body, slidably disposed in the sleeve; a valve spring, disposed in the sleeve to bias the spool valve body in an axis direction; a check valve, disposed between a tip end (bottom wall) of the valve body and an end (bottom end) of the sleeve; a fixing member, abutting against the other end (opening end) of the sleeve; and an elastic member (a seal member forming an O-ring) disposed between the check valve and the bottom wall of the valve body (see, for example, Patent Document 1).

(4) In the hydraulic control valve, the check valve includes a cylindrical body part, a ball valve body movably disposed in the body part in the axis direction, a coil spring biasing the ball valve body in a valve closing direction, and a filter caulked and fixed to the body part. In addition, in the sleeve, an end in the axis direction is held by an elastic member (seal member) disposed between a tip end (bottom wall) of the valve body and a body part of the check valve, and the other end in the axis direction is held by the fixing member having an annular shape and pressed against the inner wall surface of the valve body.

(5) Here, the seal member applying a biasing force to the end of the sleeve has a structure applying the biasing force via the body part of the check valve, and the filter has a structure of being fixed to the body part by caulking. Therefore, the number of components is large, the structure is complex, and the assembling process is complicated. In addition, the fixing member holding the other end of the sleeve is fixed by being pressed against the inner wall surface of the valve body. Therefore, due to the impact or changes over time, etc., resulting from the reciprocation of the spool valve body, the fixing member may fall off.

(6) In addition, as another fluid control valve unit, a control valve including the following is known: a linking bolt, having an insertion hole with an axis as the center; a spool and retainer, accommodated in the linking bolt; a spool spring, disposed between a spool and the retainer; and a check valve, connected to the retainer by sandwiching an O-ring, the check valve including a ball holder, a conical oil filter provided at the ball holder, a check ball, and a check spring (see, for example, Patent Document 2).

(7) In the control valve, the shapes of the ball holder and the oil filter are complicated, and the O-ring is adopted as an elastic member applying a biasing force in the axis direction. Therefore, the number of components is large, the structure is complex, and the assembling process is complicated.

Prior Art Document(s)

Patent Document(s)

(8) [Patent Document 1] Japanese Laid-open No. 2015-124643. [Patent Document 2] Japanese Laid-open No. 2017-122418.

(9) The invention provides a fluid control valve unit with which the structure can be simplified, the number of components can be reduced, the cost can be reduced, the assembling process can be simplified, a foreign matter can be prevented from entering, and a sleeve can be held at a predetermined position to ensure functional reliability, as well as a valve timing change device using the fluid control valve unit.

SUMMARY

(10) A fluid control valve unit according to an aspect of the invention includes: a sleeve, having a bottomed cylindrical shape, having: a fluid control valve, including: an inlet; a communication port in communication with outside; a bottom end; and an opening end, and defining an axis; and a spool slidably disposed in the sleeve to open and close the communication port; a passage member, having a cylindrical shape and provided with: an inner circumferential surface, fit with the sleeve; a passage of a fluid, leading to the communication port; an annular receiving part, facing the bottom end in a direction of the axis; and an opening part formed in adjacency with the annular receiving part and allowing the fluid to flow in; a filter, disposed between the bottom end and the annular receiving part; and a stopper, receiving the opening end of the sleeve accommodated in the passage member. The filter is integrally formed by a metal plate spring member to include an annular plate

part, a filter part surrounded by the annular plate part, and a plate spring piece extending from the annular plate part and applying a biasing force in the direction of the axis.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. **1** is a schematic view illustrating a configuration of an engine to which a valve timing change device is applied, the valve timing change device including a fluid control valve unit according to an embodiment.

(2) FIG. **2** is an exploded perspective view, when viewed from the oblique front of a side opposite to a camshaft, illustrating an electromagnetic actuator, a fastening bolt in which a fluid control valve is built, the valve timing change device, and the camshaft in the configuration shown in FIG. **1**.

(3) FIG. **3** is an exploded perspective view, when viewed from the oblique rear of a side of the camshaft, illustrating the electromagnetic actuator, the fastening bolt in which the fluid control valve is built, the valve timing change device, and the camshaft in the configuration shown in FIG. **1**.

(4) FIG. **4** is an exploded perspective view, when viewed from the oblique front of the side opposite to the camshaft, illustrating a housing rotor, a vane rotor, a rotation biasing spring, and the camshaft included in the valve timing change device according to the invention.

(5) FIG. **5** is an exploded perspective view, when viewed from the oblique rear on the side of the camshaft, illustrating the housing rotor, the vane rotor, the rotation biasing spring, and the camshaft included in the valve timing change device according to the invention.

(6) FIG. **6** is a cross-sectional view illustrating a locked state in which a lock mechanism is operated in a state in which the valve timing change device of the invention is fastened and fixed to the camshaft by using a fastening bolt.

(7) FIG. **7** is a cross-sectional view illustrating a passage of a region around the fluid control valve unit in the state in which the valve timing change device of the invention is fastened and fixed to the camshaft by using the fastening bolt.

(8) FIG. **8** is an appearance perspective view illustrating a fluid control valve unit according to an embodiment.

(9) FIG. **9** is an exploded perspective view, when viewed from the oblique front of the side opposite to the camshaft to which the valve timing change device is assembled, illustrating the fluid control valve unit according to an embodiment.

(10) FIG. **10** is an exploded perspective view, when viewed from the oblique rear of the side of the camshaft to which the valve timing change device is installed, illustrating the fluid control valve unit according to an embodiment.

(11) FIG. **11** is a perspective view, when viewed from the side of an annular plate part, illustrating a filter included in the fluid control valve unit according to the invention.

(12) FIG. **12** is a perspective view, when viewed from the side of multiple plate spring pieces, illustrating the filter included in the fluid control valve unit according to an embodiment.

(13) FIG. **13** is a plan view illustrating a state before a punching and bending process is applied to a filter included in the fluid control unit according to an embodiment from a spring steel plate as a metal spring material.

(14) FIG. **14** is a perspective view illustrating the filter disposed in adjacency with a bottom end of a sleeve in the fluid control valve unit according to an embodiment.

(15) FIG. **15** is an exploded perspective view illustrating the sleeve and the filter in the fluid control valve unit according to an embodiment.

(16) FIG. **16** is a cross-sectional view, illustrating the fluid control valve unit according to an

embodiment, in a region of a discharge passage able to communicate with a first communication port and a second communication port.
(17) FIG. **17** is a cross-sectional view, illustrating the fluid control valve unit according to an embodiment, in a region of a discharge passage in communication with a region in which a biasing spring biasing a spool is disposed.
(18) FIG. **18** is a cross-sectional view, illustrating the fluid control valve unit according to an embodiment, in a region of a cutout passage and an inlet (supply port).
(19) FIG. **19** is a cross-sectional view, illustrating the fluid control valve unit according to an embodiment, in a region of a first communication port (retard port) and a second communication port (advance port).
(20) FIG. **20** is a perspective cross-sectional view illustrating the spool included in the fluid control valve unit according to the invention and taken along a surface including an axis.
(21) FIG. **21** is a cross-sectional view illustrating a state in which the vane rotor is locked at an intermediate position with respect to the housing rotor.
(22) FIG. **22** is a cross-sectional view illustrating a state in which the vane rotor is located at a most retard position with respect to the housing rotor.
(23) FIG. **23** is a cross-sectional view illustrating a state in which the vane rotor is located at a most advance position with respect to the housing rotor.
(24) FIG. **24** is a schematic view illustrating a relationship between the spool of the fluid control valve, the retard port, the advance port, and a hydraulic oil flow in a retard chamber and an advance chamber when the camshaft receives a reverse torque in a retard mode.
(25) FIG. **25** is a schematic view illustrating a relationship between the spool of the fluid control valve, the retard port, the advance port, and the hydraulic oil flow in the retard chamber and the advance chamber when the camshaft receives a forward torque in the retard mode.
(26) FIG. **26** is a schematic view illustrating a relationship between the spool of the fluid control valve, the retard port, the advance port, and the hydraulic oil flow in the retard chamber and the advance chamber when the camshaft receives a reverse torque in an advance mode.
(27) FIG. **27** is a schematic view illustrating a relationship between the spool of the fluid control valve, the retard port, the advance port, and the hydraulic oil flow in the retard chamber and the advance chamber when the camshaft receives a forward torque in the advance mode.
(28) FIG. **28** is a schematic view illustrating a relationship between the spool of the fluid control valve, the retard port, the advance port, and the hydraulic oil flow in the retard chamber and the advance chamber when the camshaft receives a reverse torque in a neutral hold mode.
(29) FIG. **29** is a schematic view illustrating a relationship between the spool of the fluid control valve, the retard port, the advance port, and the hydraulic oil flow in the retard chamber and the advance chamber when the camshaft receives a forward torque in the neutral hold mode.
(30) FIG. **30** is a perspective view illustrating another embodiment of a filter included in the fluid control valve unit according to the invention.
DESCRIPTION OF THE EMBODIMENTS
(31) A fluid control valve unit according to an aspect of the invention includes: a fluid control valve, including: an inlet; a communication port in communication with outside; a sleeve having a bottomed cylindrical shape having a bottom end and an opening end and defining an axis; and a spool slidably disposed in the sleeve to open and close the communication port; a passage member, having a cylindrical shape and provided with: an inner circumferential surface, fit with the sleeve; a passage of a fluid, leading to the communication port; an annular receiving part, facing the bottom end in a direction of the axis; and an opening part formed in adjacency with the annular receiving part and allowing the fluid to flow in; a filter, disposed between the bottom end and the annular receiving part; and a stopper, receiving the opening end of the sleeve accommodated in the passage member. The filter is integrally formed by a metal plate spring member to include an annular plate part, a filter part surrounded by the annular plate part, and a plate spring piece extending from the

annular plate part and applying a biasing force in the direction of the axis.
(32) In the fluid control valve unit, it may also be configured that the passage member has an annular groove adjacent to the opening end in the direction of the axis and recessed with respect to the inner circumferential surface, and the stopper is a snap ring fit into the annular groove to detachably receive the spool.
(33) In the fluid control valve unit, it may also be configured that the filter is disposed so that the annular plate part abuts against the annular receiving part, and the plate spring piece abuts against the bottom end.
(34) In the fluid control valve unit, it may also be configured that the filter includes multiple micro pores formed through an etching process.
(35) In the fluid control valve unit, it may also be configured that the plate spring piece is formed to extend from an outer edge of the annular plate part and be bent.
(36) In the fluid control valve unit, it may also be configured that the sleeve includes, in a region of the bottom end, a cutout passage formed on an outer wall to supply the fluid passing through the filter to the inlet; and an abutting part abutting against the plate spring piece of the filter.
(37) In the fluid control valve unit, it may also be configured that the cutout passage includes a first cutout passage and a second cutout passage formed as separated from each other about the axis, the abutting part includes: a narrow abutting part located between the first cutout passage and the second cutout passage; and a wide abutting part away from the cutout passage, and the plate spring piece includes a first plate spring piece, a second plate spring piece, and a third plate spring piece disposed at equal intervals in a circumferential direction of the annular plate part.
(38) In the fluid control valve unit, it may also be configured that the first plate spring piece includes a pair of clamping pieces abutting against the narrow abutting part and sandwiching the narrow abutting part.
(39) In the fluid control valve unit, it may also be configured that the fluid control valve includes a check valve disposed on an inner side of the sleeve to allow the fluid to flow in from the inlet.
(40) In the fluid control valve unit, it may also be configured that the snap ring has a discharge port able to discharge the fluid flowing through a discharge passage formed in the passage member.
(41) In the fluid control valve unit, it may also be configured that the snap ring is formed in a plate shape expanding in a direction perpendicular to the axis.
(42) In the fluid control valve unit, it may also be configured that the sleeve includes, as the communication port, a first communication port and a second communication port located on two sides sandwiching the inlet in the direction of the axis, and the spool includes: a rod, reciprocating in the sleeve; a first valve part, provided at the rod to open and close a passage between the inlet and the first communication port; a second valve part, provided at the rod to open and close a passage between the inlet and the second communication port; and a biasing spring, applying a biasing force in a direction that brings the first valve part to abut against the snap ring.
(43) In the fluid control valve unit, it may also be configured that the spool includes a compression spring disposed between the first valve part and the second valve part, the first valve part includes: a first fixed part, having a first land able to block the first communication port and a first inner passage formed on an inner side of the first land, and being fixed to the rod; and a first movable part, having a first cover part opening and closing the first inner passage and movably supported along the rod, a second fixed part, having a second land able to block the second communication port and a second inner passage formed on an inner side of the second land, and being fixed to the rod; and a second movable part, having a second cover part opening and closing the second inner passage and movably supported along the rod, and the compression spring is disposed to apply a biasing force to close the first cover part and close the second cover part.
(44) A valve timing change device according another aspect of the invention is a valve timing change device changing an opening/closing timing of an intake air valve or an exhaust air valve driven by a camshaft. The valve timing change device comprising: a housing rotor, rotating

coaxially with the camshaft; a vane rotor, cooperating with the housing rotor to define an advance chamber and a retard chamber and integrally rotating with the camshaft; the fluid control valve unit, forming a configuration in which the sleeve includes the inlet, the first communication port, and the second communication port, and the spool includes the first valve part and the second valve part, and configured to control supplying and discharging of hydraulic oil with respect to the advance chamber and the retard chamber. The inlet of the fluid control valve unit is a supply port to which the hydraulic oil is supplied. The first communication port of the fluid control valve unit is a retard port in communication with the retard chamber. The second communication port of the fluid control valve unit is an advance port in communication with the advance chamber. In the valve timing change device, it may also be configured that the valve timing change device includes: a fastening bolt fastening the vane rotor to the camshaft, and the fastening bolt serves as the passage member of the fluid control valve unit.
(45) In the valve timing change device, it may also be configured that the fluid control valve of the fluid control valve unit is a torque-driven and hydraulic pressure-driven type fluid control valve reciprocating the hydraulic oil between the retard chamber and the advance chamber by using a variable torque received by the camshaft and being able to discharge a portion of the hydraulic oil that is supplied.
(46) In the valve timing change device, it may also be configured that in a state of being positioned in a retard mode in which the first valve part is opened and the second valve part is closed, when the camshaft receives a reverse torque, the second valve part is opened to allow a flow of the hydraulic oil from the advance port toward the retard port, and in a state of being positioned in an advance mode in which the first valve part is closed and the second valve part is opened, when the camshaft receives a forward torque, the first valve part is opened to allow a flow of the hydraulic oil from the retard port toward the advance port.
(47) In the valve timing change device, it may also be configured that in a state of being positioned in a neutral hold mode in which the first valve part blocks the retard port and the second valve part blocks the advance port, reciprocation of the hydraulic oil between the retard chamber and the advance chamber is cut off.
(48) According to the fluid control valve unit with the above configuration, the structure can be simplified, the number of components can be reduced, the cost can be reduced, and the assembling process can be simplified, and a foreign matter can be prevented from entering, and the sleeve can be held at a predetermined position to ensure functional reliability. In addition, according to the valve timing change device including the fluid control valve unit with the above configuration, the device size can be reduced, the sleeve can be held at a predetermined position to ensure reliability, and the desired operation can be attained.
(49) In the following, the embodiments of the invention will be described with reference to the accompanying drawings.
(50) As shown in FIG. **1**, a valve timing change device M including a fluid control valve unit U according to an embodiment is installed to a camshaft **1** of an internal combustion engine, and changes an opening/closing period of an intake air valve or an exhaust air valve driven by the camshaft **1**. That is, the valve timing change device M changes a valve timing.
(51) The internal combustion engine includes: the camshaft **1**, driving to open/close the intake air valve or the exhaust air valve; an oil pan **2**, storing hydraulic oil; a supply passage **3**, supplying the hydraulic oil as a fluid in the oil pan **2** toward the camshaft **1**; an oil pump **4**, provided midway of the supply passage **3**, and attracting, pressurizing, and discharging the hydraulic oil; a discharge passage **5**, bringing the hydraulic oil discharged from a fluid control unit U back to the oil pan **2**; a chain cover **6**, covering the valve timing change device M; and an electromagnetic actuator **7**, fixed to the chain cover **6**.
(52) As shown in FIGS. **1** to **7**, the camshaft **1** rotates in a direction CR with an axis S as the center, and includes a fitting shaft part **1***a*, passages **1***b* and **1***c*, a female screw part **1***d*, and a fitting hole **1***e*

to be fit with a positioning pin P.

(53) The supply passage **3** is formed in a cylinder block and a cylinder head, etc., of the internal combustion engine.

(54) The discharge passage **5** is defined between the cylinder block as well as the cylinder head and the chain cover **6** of the internal combustion engine, and brings extra hydraulic oil discharged from the fluid control unit U back to the oil pan **2**.

(55) The electromagnetic actuator **7** is fixed to the chain cover **6** and, as shown in FIG. **3**, includes a driving shaft **7***a* moving in the direction of the axis and an excitation coil (not shown) driving the driving shaft **7***a*.

(56) As shown in FIGS. **2** to **7**, the valve timing change device M includes a housing rotor **10**, a vane rotor **20**, a rotation biasing spring **30**, a lock mechanism **40**, and the fluid control valve unit U.

(57) The fluid control valve unit U includes a fastening bolt **50** as a passage member, a filter **60**, a fluid control valve V, and a snap ring **110** as a stopper.

(58) The fluid control valve V serves to switch the passage to control the flow of the hydraulic oil, and includes a sleeve **70**, a spool **80**, a biasing spring **90**, and a check valve **100**.

(59) The housing rotor **10** is rotatably supported on the axis S of the camshaft **1** and linked with the rotation of a crankshaft via a chain, and transmits a rotational driving force of a crankshaft to camshaft **1** via the vane rotor **20**.

(60) As shown in FIGS. **4** to **7**, the housing rotor **10** has a two-part structure formed by a first housing **11** having a disk shape and a second housing **12** having a bottomed cylindrical shape and combined with the first housing **11**. In addition, the housing rotor **10** accommodates the vane rotor **20** to be relatively rotatable within an angle range between the most retard position and the most advance position of the vane rotor **20**, and cooperates with the vane rotor **20** to define an advance chamber AC and a retard angle chamber RC.

(61) The first housing **11** includes a sprocket **11***a*, a fitting hole **11***b*, an inner wall surface **11***c*, a lock hole **11***d*, a concave part **11***e* formed continuously with the lock hole **11***d*, three circular holes **11***f* through which screws b pass, and a positioning hole **11***g* fit with a positioning pin P.sub.2. The fitting hole **11***b* is rotatably fit with a fitting shaft part **1***a* of the camshaft **1**. The inner wall surface **11***c* slidably contacts a back surface **24** of the vane rotor **20**. A lock pin **41** of the lock mechanism **40** fits into the lock hole **11***d* with a small gap. The concave part **11***e* is formed on the circumference of the lock hole **11***d* to guide the hydraulic oil to a tip pressure-receiving part **41***a* of the lock pin **41** fit into the lock hole **11***d*.

(62) As shown in FIGS. **4** to **7**, the second housing **12** includes a cylindrical wall **12***a*, a front wall **12***d*, an opening part **12***c*, three screw holes **12***d* into which the screws b are screwed, three shoe parts **12***e*, a hooking groove **12***f*, a concave part **12***g*, an annular bonding part **12***h* bonded to the inner wall surface **11***c* of the first housing **11**, and a positioning hole **12***i* into which the positioning pin P.sub.2 is fit.

(63) The opening part **12***c* forms a circular hole with the axis S as the center, so that the fastening bolt **50** passes through.

(64) On the inner side of the front wall **12***b*, the three shoe parts **12***e* protrude from the cylindrical wall **12***a* toward the center, and are formed by being disposed at equal intervals in the circumferential direction.

(65) One of the shoe parts **12***e* abuts against a vane part **22** of the vane rotor **20** to define a maximum retard position, and another one of the shoe parts **12***e* abuts against the vane part **22** to define a maximum advance position.

(66) The hooking groove **12***f* is formed by cutting out a part of the opening **12***c* to lock a first end **32** of the rotation biasing spring **30**. The concave part **12***g* accommodates a portion of the coil part **31** of the rotation biasing spring **30**.

(67) The vane rotor **20** is disposed on the inner side of the housing rotor **10**, cooperates with the housing rotor **10** to define the advance chamber AC and the retard chamber RC, is fixed to the

camshaft **1** by the fastening bolt **50** by sandwiching a washer W, and integrally rotates with the camshaft **1**.
(68) As shown in FIGS. **4** to **7**, the vane rotor **20** includes a hub part **21**, three vane parts **22**, a front surface **23**, an annular concave part **23***a*, a hooking groove **23***b*, a back surface **24**, a fitting hole **25**, a concave part **26**, a communication path **27**, a retard passage **28**, and an advance passage **29**.
(69) The vane parts **22** cooperate with the shoe parts **12***e* of the housing rotor **10** to define the advance chamber AC and the retard chamber RC. The front surface **23** is slidably disposed on and brought into contact with the inner wall surface of the front wall **12***b* of the second housing **12**. The annular concave part **23***a* is formed to cut out the front surface in an annular shape to accommodate a portion of the coil part **31** of the rotation biasing spring **30**. The hooking groove **23***b* is formed by cutting out a portion of the front surface **23**, so as to hook a second end **33** of the rotation biasing spring **30**.
(70) The back surface **24** is formed on a plane perpendicular to the axis S, bonded to the end surface of the camshaft **1**, and is slidably disposed on and brought into contact with the inner wall surface **11***c* of the first housing **11**. In addition, on the back surface **24**, a fitting hole **24***a* into which the positioning pin P is fit is provided. The positioning pin P is to be installed to the fitting hole **1***e* of the camshaft **1**. The fitting hole **25** is formed with an inner diameter dimension with which a cylindrical part **50***a* of the fastening bolt **50** is fit closely.
(71) As shown in FIGS. **5** and **6**, the concave part **26** is formed to accommodate the lock mechanism **40** in one of the vane parts **22**, and includes a receiving part **26***a* and a communication path **26***b*. The receiving part **26***a* receives the biasing spring **42** included in the lock mechanism **40**. The communication path **26***b* is in communication with the outside of the vane rotor **20**.
(72) The communication path **27** is formed by an annular communication path **27***a* and a linear communication path **27***b*, and cooperates with the end surface of the camshaft **1** and the inner wall surface **11***c* of the housing rotor **10** to supply the hydraulic oil toward the lock mechanism **40** and discharge the hydraulic oil from the lock mechanism **40**. That is, the communication path **27** serves to, with respect to the lock mechanism **40**, supply the hydraulic oil guided through a through path **54** of the fastening bolt **50** on the side upstream of the fluid control valve V in the flowing direction of the supplied hydraulic oil to release locking, and serves to discharge the hydraulic oil at the time of locking. Since the communication path **27** is formed on the back surface **24** of the vane rotor **20**, processing is easy to perform, and a lubrication effect is provided in the sliding region of the inner wall surface **11***c*.
(73) The retard passage **28** supplies and discharges hydraulic oil with respect to the retard chamber RC. As shown in FIG. **22**, the retard passage **28** is formed by an annular groove **28***a* formed on the inner circumferential surface of the fitting hole **25**, and a through path **28***b* radially penetrating through the hub part **21** from the annular groove **28***a*.
(74) The advance passage **29** supplies and discharges hydraulic oil with respect to the advance chamber AC. As shown in FIG. **23**, the advance passage **29** is formed by an annular groove **29***a* formed on the inner circumferential surface of the fitting hole **25**, and a through path **29***b* radially penetrating through the hub part **21** from the annular groove **29***a*.
(75) As shown in FIGS. **4** to **7**, the rotation biasing spring **30** is a coil spring having a coil part **31**, a first end **32**, and a second end **33**.
(76) In addition, in the rotation biasing spring **30**, the coil part **31** is accommodated in the annular concave part **23***a* of the vane rotor **20** and the concave part **12***g* of the housing rotor **10**, the first end **32** is hooked to the hook groove **12***f* of the housing rotor **10**, and the second end **33** is hooked to the hook groove **23***b* of the vane rotor **20**. Accordingly, the rotation biasing spring **30** rotationally biases the vane rotor **20** in the advance direction with respect to the housing rotor **10**.
(77) In this way, by adopting the rotation biasing spring **30** biasing in the advance direction, the responsiveness can be facilitated by assisting an operation torque at the time of advancing. By setting the load of the rotation biasing spring **30** so that the difference between the operation torque

and the load torque is substantially equal at the time of advancing and retarding, the controllability can be facilitated.
(78) As shown in FIG. **6**, the lock mechanism **40** includes the lock pin **41**, the biasing spring **42**, and a cylindrical holder **43**. In addition, as shown in FIG. **21**, the lock mechanism **40** locks the vane rotor **20** at an intermediate position between the most retard position and the most advance position with respect to the housing rotor **10**.
(79) The lock pin **41** forms a substantially cylindrically columnar part, and has the tip pressure-receiving part **41***a*. In addition, the lock pin **41** is retractably held in the direction of the axis S with respect to the back surface **24** of the vane rotor **20**, so as to be fittable into the lock hole **11***d* of the housing rotor **10**. The biasing spring **42** biases the lock pin **41** in a protruding direction. The cylindrical holder **43** is fit and fixed to the concave part **26** of the vane rotor **20**, so as to reciprocally hold the lock pin **41** biased by the biasing spring **42**. In addition, as shown in FIGS. and **6**, the cylindrical holder **43** is disposed to be immersed from the back surface **24** of the vane rotor **20**, so as to define an annular oil reservoir C in communication with the linear communication path **27***b* on the circumference of the lock pin **41**. By providing the annular oil reservoir C, the hydraulic oil can be filled around the lock pin **41** to smoothly release locking.
(80) In addition, by starting the engine, the hydraulic oil pressurized by the oil pump **4** is guided to the lock mechanism **40** through the passages **1***b*, **1***c* of the camshaft **1**, the opening part **52** of the fastening bolt **50**, the filter part **62** of the filter **60**, a gap passage Cp defined by a cutout passage **71***a*, the through path **54** of the fastening bolt **50**, and the communication path **27** formed on the back surface **24** of the vane rotor **20** and the annular oil reservoir C. When the hydraulic pressure applied to the tip pressure-receiving part **41***a* of the lock pin **41** increases, the lock pin **41** is detached from the lock hole **11***d* to remove locking.
(81) Meanwhile, by stopping the engine, when the hydraulic pressure of the hydraulic oil that is supplied decreases, the hydraulic oil acting on the lock pin **41** flows out through the communication path **27** and the through path **54**, the gap passage Cp, the filter **62**, the opening part **52**, and the paths **1***c*, **1***b*, and the hydraulic pressure pressurizing the lock pin **41** decreases. Then, the lock pin **41** is biased by the biasing spring **42** to be fit into the lock hole **11***d* of the housing rotor **10**, and the vane rotor **20** is locked at the intermediate position with respect to the housing rotor **10**.
(82) As shown in FIGS. **2**, **3**, and **6** to **10**, the fastening bolt **50** includes the cylindrical part **50***a* with the axis S as the center, an inner circumferential surface **51** to which the fluid control valve V is fit, the opening part **52**, the annular receiving part **53**, the through path **54**, the retard passage **55**, the advance passage **56**, a flanged head part **57**, a male screw part **58**, a positioning concave part **59***a*, and an annular groove **59***b*.
(83) The cylindrical part **50***a* is formed with an outer diameter dimension closely fit with the fitting hole **25** of the vane rotor **20**. The inner circumferential surface **51** forms a cylindrical surface with the axis S as the center, so that the sleeve **70** is fit.
(84) The opening part **52** is adjacent to the annular receiving part **53** in the direction of the axis S to be formed in a circular hole having a diameter smaller than the inner circumferential surface **51**, and serves as a passage for the hydraulic oil to flow into the fastening bolt **50** on the upstream side with respect to the filter **60**.
(85) The annular receiving part **53** is an annular concave part and has a receiving surface formed as a planar surface perpendicular to the axis S, so as to face a bottom end **70***a* of the sleeve **70** on the inner side with respect to the opening part **52** in the direction of the axis S. The through path **54** guides the hydraulic oil in or out with respect to the lock mechanism **40**, and penetrates in a radial direction perpendicular to the axis S in the cylindrical part **50***a*.
(86) The retard passage **55** penetrates in a radial direction perpendicular to the axis S in the cylindrical part **50***a* to be in communication with the retard passage **28** of the vane rotor **20**.
(87) The advance passage **56** penetrates in a radial direction perpendicular to the axis S in the

cylindrical part **50***a* to be in communication with the advance passage **29** of the vane rotor **20**.
(88) The flanged head part **57** abuts against the front surface **23** of the vane rotor **20** to sandwich the washer W. The male screw part **58** is screwed with the female screw part **1***d* of the camshaft **1**.
(89) The positioning concave part **59***a* is formed to be fit with the positioning convex part **79** of the sleeve **70** included in the fluid control valve V and a fitting convex part **114** of the snap ring **110**. The annular groove **59***b* is adjacent to an opening end **70***b* of the sleeve **70** and formed to be recessed with respect to the inner circumferential surface **51**, so that the snap ring **110** is fit into the annular groove **59***b*, such that plate spring pieces **63** of the filter **60** are compressed and the snap ring **110** abuts against the opening end **70***b* of the sleeve **70** fit with the inner circumferential surface **51**.
(90) The filter **60** serves to capture a foreign matter mixed into the hydraulic oil supplied by an oil pump **74**, and is formed by using a metal plate spring material, such as a spring steel plate of stainless steel (SUS301, etc.). As shown in FIGS. **6**, **7**, **11**, and **12**, the filter **60** includes an annular plate part **61**, the filter part **62**, the plate spring piece **63**, and a curved notch part **64** formed in a root region of the plate spring piece **63**.
(91) The annular plate part **61** forms an annular plate with the axis S as the center, and serves as a seal member in close contact with the annular receiving part **53** of the fastening bolt **50**. The filter part **62** is formed by multiple micro pores **62***a* formed in a circular region surrounded by the annular plate part **61**. The pore diameter of the micro pore **62***a* is smaller than the size of the foreign matter anticipated to be mixed into the hydraulic oil, and is formed in a size so that the passage resistance does not increase.
(92) The plate spring piece **63** is formed to extend in the direction of the axis S from the outer edge of the annular plate part **61** and extend so that the tip end thereof expands toward the inner side of the axis S. That is, the plate spring piece **63** is formed to extend and curve to extend from the outer edge of the annular plate part **61**. Accordingly, the plate spring piece **63** is formed to apply a biasing force in the direction of the axis S.
(93) Here, the plate spring piece **63** is formed to include a first plate spring piece **63***a*, a second plate spring piece **63***b*, and a third plate spring piece **63***c* disposed at equal intervals (intervals of 120 degrees) in the circumferential direction of the annular plate part **61**.
(94) In addition, the first plate spring piece **63***a*, the second plate spring piece **63***b*, and the third plate spring piece **63***c* are disposed, so that one of the plate spring pieces **63***a* abuts against a narrow abutting part **70***a*.sub.4 of the sleeve **70** and the other two abut against a wide abutting part **70***a*.sub.5. The curved notch part **64** is formed in a root region in which the plate spring pieces **63** extend from the annular plate part **61**, alleviates the concentration of the stress generated in the root region when the plate spring pieces **63** are deformed elastically, and prevents cracks, etc., from occurring.
(95) Regarding the manufacture of the filter **60**, a spring steel plate of stainless steel (SUS301, etc.) with a plate thickness of about 0.2 mm to about 0.3 mm is punched out as a plate-shaped blank material Bm by using a punching machine, as shown in FIG. **13**. Here, the blank material Bm forms a shape including a disk part Bm and three expansion pieces Bm.sub.2 expanding radially from the disk part Bm.sub.1.
(96) In addition, in the blank material Bm, a resist film including multiple micro pores is coated on the inner side (a region corresponding to the filter part **62**) of an annular region on the outer side of the disk part Bm.sub.1, and an etching process (wet etching or dry etching) is applied from the outer side thereof. Accordingly, the micro pores **62***a* are formed in the filter part **62**. Here, the micro pore **62***a* is formed as a circular hole with an inner diameter of about 0.2 mm, for example. Then, when the etching process ends, the resist film is removed.
(97) Then, by using a bending machine, the expansion piece Bm.sub.2 is subjected to a bending process, and the plate spring pieces **63** (the first plate spring piece **63***a*, the second plate spring piece **63***b*, and the third plate spring piece **63**) are formed.

(98) In a state of being assembled as a fluid control valve unit U, the filter **60** is sandwiched by the bottom end **70***a* of the sleeve **70** (a narrow abutting part **70***a***4** and a wide abutting part **70***a*.sub.5 as the abutting part) and the annular receiving part **53** of the fastening bolt **50**.
(99) Accordingly, the region of the filter part **62** is positioned in a region facing the opening part **52** of the fastening bolt **50** and the cutout passage **71***a* of the sleeve **70**, and the plate spring pieces **63** (the first plate spring piece **63***a*, the second plate spring piece **63***b*, and the third plate spring piece **63***c*) apply biasing forces biasing the sleeve **70** in the direction of the axis S with respect to the fastening sleeve **50**.
(100) In this way, the filter **60** is integrally formed by a metal plate spring material to include the annular plate part **61**, the filter part **62**, and multiple plate spring pieces **63**. Therefore, compared with the filter formed by assembling various parts and materials, the structure can be simplified, and the cost can be reduced.
(101) Here, the plate spring piece **63** is formed as multiple (three) plate spring pieces, i.e., the first plate spring piece **63***a*, the second plate spring piece **63***b*, and the third plate spring piece **63***c*. Therefore, the structure can be simplified, the spring force can be easily generated, and the entire region of the annular plate part **61** in the circumferential direction can be in close contact with the annular receiving part **53**.
(102) The fluid control valve V serves to switch the passages to supply or discharge the hydraulic oil with respect to the advance chamber AC and the retard chamber RC. As shown in FIGS. **9**, **10**, and **16** to **20**, the fluid control valve V includes the sleeve **70**, the spool **80**, the biasing spring **90**, and the check valve **100**.
(103) The sleeve **70** is formed by using aluminum or other metal materials, and is formed in a bottomed cylindrical shape defining the axis S. The sleeve **70** includes the bottom end **70***a* and the opening end **70***b* forming two ends in the direction of the axis S, the outer wall **71**, the cutout passages **71***a*, **71***b*, **71***c*, a first communication path **71***d* and through paths **71***d*.sub.1, **71***d*.sub.2 forming a portion of the discharge passage, a second communication path **71***e* and a through path **71***e*.sub.1 forming a portion of the discharge passage, a fitting hole **71***f*, a fitting pin **71***g*, a communication concave part **71***h* forming a portion of the discharge passage, an inner circumferential surface **72**, annular grooves **72***a*, **72***b*, **72***c*, an opening part **73**, a supply port **74**, a retard port **75**, an advance port **76**, a stopper wall **77**, a spring receiving part **78**, and a positioning convex part **79**.
(104) As shown in FIGS. **14** and **15**, the bottom end **70***a* is formed by an end surface **70***a*.sub.1, inner wall parts **70***a*.sub.2, **70***a*.sub.3, the narrow abutting part **70***a*.sub.4, the wide abutting part **70***a*.sub.5, and two cutout parts **70***a*.sub.6.
(105) In an assembled state, the end surface **70***a*.sub.1 is formed to face a predetermined gap in the direction of the axis S with respect to the annular receiving part **53**, such as a gap comparable to the plate thickness of the filter **60**.
(106) The inner wall parts **70***a*.sub.2, **70***a*.sub.3 are formed so that the outer side surfaces of the plate spring pieces **63** are inserted into fine gaps, and position the filter **60** in a direction perpendicular to the axis S.
(107) The narrow abutting part **70***a*.sub.4 is located between the two cutout passages **71***a*, **71***a* as first and second cutout passages when viewed in the direction of the axis S, expands in a direction perpendicular to the axis S, and is formed as a projected strip-shaped planar surface forming a rectangular cross-section.
(108) In addition, the narrow abutting part **70***a***4** receives one of the first plate spring piece **63***a*, the second plate spring piece **63***b*, and the third plate spring piece **63***c*.
(109) The wide abutting part **70***a*.sub.5 is formed as a planar surface in a semi-circular shape perpendicular to the axis S on the same surface as the narrow abutting part **70***a*.sub.4 in a region away from the two cutout passages **71***a*, **71***a*.
(110) In addition, the wide abutting part **70***a*.sub.5 receives the remaining two of the first plate

spring piece **63***a*, the second plate spring piece **63***b*, and the third plate spring piece **63***c*.
(111) The two cutout parts **70***a*.sup.6 are formed to be in communication with the two cutout passages **71***a*, **71***a*, respectively, as the passages of the hydraulic oil on two sides of the narrow abutting part **70***a*.sub.4.
(112) As shown in FIGS. **9** and **16** to **19**, the opening end **70***b* is in an annular shape with the axis S as the center, forms a planar surface perpendicular to the axis S, and is formed to abut against the annular receiving part **111** of the snap ring **110**. In addition, in the state in which the sleeve **70** is assembled, the opening end **70***b* is received by the annular receiving part **111** of the snap ring **110** fit into the annular groove **59***b* of the fastening bolt **50** in the direction of the axis S.
(113) The outer wall **71** is formed as the cylindrical surface with the axis S as the center, and is closely fit with the inner circumferential surface **51** of the fastening bolt **50**.
(114) The two cutout passages **71***a* expand from the central region of the bottom end **70***a* as the outer side of the bottom wall to be bifurcated via the cutout parts **70***a***6**, are formed by cutting out portions of the outer wall **71** over regions facing the supply port **74** (a first supply port **74***a*, a second supply port **74***b*), and define the gap passage Cp together with the inner wall of the fastening bolt **50**. The cutout passage **71***b* is formed by cutting out a portion of the outer wall **71** in a region facing the retard passage **55** of the fastening bolt **50** from the retard port **75** and serves as a passage between the retard port **75** and the retard passage **55**.
(115) The cutout passage **71***c* is formed by cutting out a portion of the outer wall **71** in a region facing the advance passage **56** of the fastening bolt **50** from the advance port **76** and serves as a passage between the advance port **76** and the advance passage **56**.
(116) The communication path **71***d* is formed to expand in the direction of the axis S on the outer wall **71**, and serves as a discharge passage in cooperation with the through path **71***d*.sub.1 to communicate with the retard port **75** to be able to discharge the hydraulic oil when a first valve part **82** is closed and in cooperation with the through path **71***d***2** to communicate with the advance port **76** to discharge the hydraulic oil when a second valve part **83** is closed.
(117) The communication path **71***e* is formed to expand in the direction of the axis S on the outer wall **71** at a position away from the communication path **71***d*, and cooperates with the through path **71***e*.sub.1 to serve as a discharge passage able to discharge the hydraulic oil accumulated in the region disposed in the biasing spring **90** and serving as a breathing path.
(118) The fitting hole **71***f* is fit with the fitting pin **71***g*, and is formed as a double hole radially penetrating in the radial direction on the bottom wall of the annular groove **72***a*.
(119) The fitting pin **71***g* forms a stepped pin in which two cylindrical columns having different outer diameters are integrally formed by using iron or steel materials, is closely fit with the fitting hole **71***f*, and protrudes radially inward from the bottom surface of the annular groove **72***a* so as not to interfere with the spool **80**.
(120) The communication concave part **71***h* is formed to be in communication with the first communication path **71***d* and the second communication path **71***e* in the circumferential direction on the outer wall **71** in the vicinity of the opening part **73**.
(121) The inner circumferential surface **72** is formed in a cylindrical shape with the axis S as the center, and closely and slidably guides the first valve part **82** (a first land **82***a*.sub.1) and the second valve part **83** (a second land **83***a*.sub.1) of the spool **80**.
(122) The annular groove **72***a* is formed as a cylindrical surface annularly cut out in an annular shape so as to be recessed from the inner circumferential surface **72** in a width greater than the opening width of the supply port **74** in the direction of the axis S in a region facing the supply port **74** as an inlet, and the check valve **100** is disposed on the inner side of the annular groove **72***a*.
(123) The annular groove **72***b* is formed by being cut out in an annular shape, so as to be recessed from the inner circumferential surface **72**, in a region facing the retard port **75** as the first communication port and serves as a passage of the hydraulic oil.
(124) The annular groove **72***c* is formed by being cut out in an annular shape, so as to be recessed

from the inner circumferential surface **72**, in a region facing the advance port **76** as the second communication port, and serves as a passage of the hydraulic oil.
(125) The opening part **73** allows the rod **81** of the spool **80** to project toward the direction of the axis S.
(126) The supply port **74** serves as an inlet through which the hydraulic oil flows in as fluid, communicates with the gap passage Cp, and is disposed on the downstream side with respect to the through path **54** in the gap passage **54**.
(127) In addition, as shown in FIG. **10**, the supply port **74** includes the first supply port **74***a* as a first inlet and the second supply port **74***b* as a second inlet, the first supply port **74***a* and the second supply port **74***b* being provided to be spaced apart from each other about the axis S.
(128) The retard port **75** serves as the first communication port in communication with the outside so that the hydraulic oil as fluid passes through, communicates with the retard passage **55** of the fastening bolt **50** via the cutout passage **71***b*, and communicates with the retard chamber RC via the retard passage **28** of the vane rotor **20**.
(129) The advance port **76** serves as the second communication port in communication with the outside so that the hydraulic oil as fluid passes through, communicates with the advance passage **56** of the fastening bolt **50** via the cutout passage **71***c*, and communicates with the advance chamber AC via the advance passage **29** of the vane rotor **20**.
(130) Here, as shown in FIGS. **18** and **19**, the retard port **75** and the advance port **76** are disposed to be located on two sides sandwiching the supply port **74** in the direction of the axis S. That is, the communication port communicating with the outside so that the fluid passes through includes the first communication port (the retard port **75**) and the second communication port (the advance port **76**) located on two sides sandwiching the inlet (the supply port **74**) in the direction of the axis S. The stopper wall **77** receives an end surface **83***a*.sup.2 of the second valve part **83** of the spool **80**, and stops the spool **80** at the innermost position corresponding to the advance mode.
(131) The spring receiving part **78** receives the end of the biasing spring **90**.
(132) The positioning convex part **79** is fit with the positioning concave part **59***a* of the fastening bolt **50**, so as to position the sleeve **70** at a predetermined position about the axis S with respect to the fastening bolt **50**, when the sleeve **70** is fit with the inner circumferential surface **51** of the fastening bolt **50**.
(133) As described above, the discharge passage (the first communication path **71***d*, the second communication path **71***e*, the communication concave part **71***h*) through which the hydraulic oil discharged to the outside passes is formed on the outer wall **71** in the sleeve **70**. Therefore, it is not required to provide a discharge passage in the fastening bolt **50** as a passage member. Consequently, the fluid control valve V including the sleeve **70** can be applied to various conventional passage members.
(134) As shown in FIGS. **16** to **20**, the spool **80** is disposed on the inner side of the sleeve **70** to be slidable on the inner circumferential surface **72**, and includes the rod **81** expanding in the direction of the axis S, the first valve part **82** and the second valve part **83** provided at the rod **81**, and a compression spring **84** disposed between the first valve part **82** and the second valve part **83**. The rod **81** is formed to expand in the direction of the axis S and includes an end **81***a* exposed to the outside. The driving shaft **7***a* of the electromagnetic actuator **7** is engaged with the end **81***a*, and a driving force is applied against the biasing force of the biasing spring **90**.
(135) The first valve part **82** serves to open/close a passage between the supply port **74** and the retard port **75**, and includes a first fixed part **82***a* and a first movable part **82***b*. The first fixed part **82***a* is fixed to the rod **81**. The first movable part **82***b* is movably supported along the rod **81** and biased by the compression spring **84**.
(136) The first fixed part **82***a* includes a first land **82***a*.sub.1, an end surface **82***a*.sub.2, a first inner passage **82***a*.sub.3, and an end surface **82***a*.sub.4. The first land **82***a*.sub.1 slides closely on the inner circumferential surface **72**.

(137) The first land **82***a*.sub.1 is a cylindrical surface having an outer diameter the same as or slightly smaller than the inner diameter of the inner circumferential surface **72**, with the axis S as the center, and having a width dimension blocking the retard port **75**, and opens or blocks the retard port **75**.
(138) The first movable part **82***b* cooperates with the compression spring **84** to serve as a check valve, and includes a first fitting part **82***b*.sub.1 and a first cover part **82***b*.sub.2. The first fitting part **82***b*.sub.1 is slidably fit with the rod **81**. The first cover part **82***b*.sub.2 detachably abuts against the end surface **82***a*.sub.4, so as to open/close the first inner passage **82***a*.sub.3.
(139) The second valve part **83** serves to open/close a passage between the supply port **74** and the advance port **76**, and includes a second fixed part **83***a* and a second movable part **83***b*. The second fixed part **83***a* is fixed to the rod **81**. The second movable part **83***b* is movably supported along the rod **81** and biased by the compression spring **84**.
(140) The second fixed part **83***a* includes a second land **83***a*.sub.1, an end surface **83***a*.sub.2, a second inner passage **83***a*.sub.3, and an end surface **83***a*.sub.4. The second land **83***a*.sub.1 slides closely on the inner circumferential surface **72**.
(141) The second land **83***a*.sub.1 is a cylindrical surface having an outer diameter the same as or slightly smaller than the inner diameter of the inner circumferential surface **72**, with the axis S as the center, and having a width dimension blocking the advance port **76**, and opens or blocks the retard advance**76**.
(142) The second movable part **83***b* cooperates with the compression spring **84** to serve as a check valve, and includes a second fitting part **83***b*.sub.1 and a second cover part **83***b*.sub.2. The second fitting part **83***b*.sub.1 is slidably fit with the rod **81**. The second cover part **83***b*.sub.2 detachably abuts against the end surface **83***a*.sub.4, so as to open/close the second inner passage **83***a*.sub.3.
(143) The compression spring **84** is a compression-type coil spring and disposed between the first movable part **82***b* of the first valve part **82** and the second movable part **83***b* of the second valve part **83**, and applies a biasing force so that the first cover part **82***b*.sub.2 blocks the first inner passage **82***a*.sub.3 and the second cover part **83***b*.sub.2 blocks the second inner passage **83***a*.sub.3.
(144) Here, the relationship between the biasing force of the compression spring **84** and the passage resistance of the through paths **71***d*.sub.1 and **71***d*.sub.2 forming a portion of the discharge passage is described.
(145) In the valve-closed state of the first valve part **82**, when the pressure of the hydraulic oil flowing in from the retard port **75** is large, the first cover part **82***b*.sub.2 is opened, the hydraulic oil discharged from the discharge passage (the through path **71***d*.sub.1) decreases, and the hydraulic oil actively flows toward the side of the advance port **76**; meanwhile, when the pressure of the hydraulic oil flowing in from the retard port **75** is small, the first cover part **82***b*.sub.2 is closed, and the hydraulic oil is actively discharged from the discharge passage (the through path **71***d*.sub.1).
(146) In addition, in the valve-closed state of the second valve part **83**, when the pressure of the hydraulic oil flowing in from the advance port **76** is large, the second cover part **83***b*.sub.2 is opened, the hydraulic oil discharged from the discharge passage (the through path **71***d*.sub.2) decreases, and the hydraulic oil actively flows toward the side of the retard port **75**; meanwhile, when the pressure of the hydraulic oil flowing in from the advance port **76** is small, the second cover part **83***b*.sub.2 is closed, the hydraulic oil is actively discharged from the discharge passage (the through path **71***d*.sub.2).
(147) To perform the operation, the biasing force of the compression spring **84** and the passage resistance of the through paths **71***d*.sub.1, **71***d*.sub.2 are set.
(148) The biasing spring **90** is a compression type coil spring. As shown in FIGS. **16** to **19**, the biasing spring **90** is assembled so that an end abuts against the end surface **83***a*.sub.2 of the spool **80**, and the other end abuts against the spring receiving part **78** of the sleeve **70**. In addition, in the inactive state, the biasing spring **90** applies a biasing force that stops the spool **80** at a position corresponding to the retard mode, that is, an inactive position where the end surface **82***a*.sub.2 of

the spool **80** abuts against protrusive receiving parts **112**, **113** of the snap ring **110**.
(149) The check valve **100** is a C-shaped spring in which an elongated plate spring formed by spring steel is curved into an annular shape to make two ends face each other to form a notch part of a predetermined gap, and which is curved in advance to form an outer diameter greater than the inner diameter of the annular groove **72***a*.
(150) In addition, the check valve **100** is disposed at the annular groove **72***a* of the sleeve **70** in a diameter-reducible manner, so that the fitting pin **71***g* is located at the gap of the notch part, and serves as a check valve that only allows the supplied hydraulic oil to flow in through the supply port **74** of the sleeve **70**.
(151) It is noted that, the check valve **100** is set with a valve-opening property, so that, after the supplied hydraulic oil flows into the fluid control valve V through the supply port **74** and the gap passage Cp defined by the passages **1***b*, **1***c*, the opening part **52**, the filter part **62**, the cutout part **70***a***6**, and the cutout passage **71***a* and is supplied to the retard chamber CR from the retard port **75** or the advance chamber AC from the advance port **76**, when the hydraulic pressure of the hydraulic oil filled in the through path **54** and the communication path **27** reaches a hydraulic pressure able to release the lock mechanism **40**, locking is released.
(152) The snap ring **110** is fit into the annular groove **59***b* of the fastening bolt **50** and is formed by spring steel, etc. As shown in FIGS. **8** to **10**, the snap ring **110** forms a plate shape expanding in a direction perpendicular to the axis S and is formed in a substantially C shape having a notch part **110***a* of a predetermined gap.
(153) In addition, the snap ring **110** includes multiple (five in the embodiment) protrusive receiving parts **112**, **113** and a fitting convex part **114**. The protrusive receiving parts **112**, **113** protrude radially inward from the annular receiving part **111** at substantially equal intervals about the axis S.
(154) The annular receiving part **111** receives the opening end **70***b* of the sleeve **70** inserted into the inner circumferential surface **51** of the fastening bolt **50**.
(155) The two protrusive receiving parts **112** and three protrusive receiving parts **113** respectively receive the opening end **70***b* of the sleeve **70** in a root side region and detachably receive the end surface **82***a* of the spool **80** in a tip end side region.
(156) The two protrusive receiving part **112** are provided in the vicinities of the two ends defining the notch part **110***a*, and formed with two circular holes **112***a*, **112***b*.
(157) The two circular holes **112***a*, **112***b* are provided for the tip of a tool (e.g., a snap ring plier) to be inserted when the snap ring **110** is assembled to the annular groove **59***b*.
(158) In the snap ring **110** with such configuration, the notch part **110***a* and the two circular holes **112***a*, **112***b* serve as discharge ports able to discharge the hydraulic oil flowing through the discharge passage (the first communication path **71***d*, the second communication path **71***e*, the communication concave part **71***h*).
(159) The fitting convex part **114** is formed to be fit with the positioning concave part **59***a* of the fastening bolt **50**, so that the discharge port (the notch part **110***a* and the two circular holes **112***a*, **112***b*) corresponds to the discharge passage (the first communication path **81***b*, the second communication path **71***e*, and the communication concave part **71***h*).
(160) That is, as shown in FIGS. **10** and **16** to **19**, in the state in which the filter **60** is fit with the bottom end **70***a* of the sleeve **70**, the fluid control valve V (the sleeve **70**, the spool **80**, the biasing spring **90**, and the check valve **100**) is fit into the inner circumferential surface **51** of the fastening bolt **50**, and the bottom end **70***a* of the sleeve **70** presses the filter **60** in the direction of the axis S, by being fit into the annular groove **59***b* of the fastening bolt **50**, the snap ring **110** receives the opening end **70***b* of the sleeve **70** accommodated in the fastening bolt **50** and detachably receives the spool **80**, and is able to discharge the hydraulic oil flowing through the discharge passage (the first communication path **71***d*, the second communication path **71***e*, and the communication concave part **71***h*) formed in the fastening bolt **50**.
(161) Consequently, in the direction of the axis S inside the fastening bolt **50**, the sleeve **70** is fixed

in a state in which the bottom end **70***a* faces the annular receiving part **53** by sandwiching the filter **60** including the plate spring pieces **63**, the opening end **70***b* abuts against the snap ring **110** fit into the annular groove **59***b* of the fastening bolt **50**, and the biasing forces of the plate spring pieces **63** are applied in the direction of the axis S.

(162) According to the above, the fluid control valve unit U includes: the fluid control valve V; the fastening bolt **50** as a cylindrical passage member; the filter **60** disposed between the bottom end **70***a* of the sleeve **70** and the annular receiving part **53** of the fastening bolt **50**; and the stopper receiving the opening end **70***b* of the sleeve **70** accommodated in the fastening bolt **50**. The filter **60** is integrally formed by using a metal plate spring material, so as to include the annular plate part **61**, the filter **62** surrounded by the annular plate part **61**, and the plate spring pieces **63** extending from the annular plate part **61** to apply biasing forces in the direction of the axis S. Accordingly, by assembling the filter **60** as one component between the annular receiving part **53** of the fastening bolt **50** and the bottom end **70***a* of the sleeve **70**, the maintaining function of applying the spring biasing force, the filter function, and the seal function can be attained, the structure can be simplified, the number of components can be reduced, the assembling process can be simplified, foreign matters can be prevented from entering, and the functional reliability can be facilitated.

(163) In particular, the filter **60** is formed by a single metal plate spring material. Therefore, compared with one formed by combining multiple components, the cost can be reduced, and the durability can be facilitated.

(164) Moreover, the fastening bolt **50** as the passage member has the annular groove **72***a* adjacent to the opening end **70***b* in the direction of the axis S and is recessed with respect to the inner circumferential surface **72**, and, as the stopper, the snap ring **110** fit into the annular groove **72***a* detachably receiving the spool **80** is adopted.

(165) Accordingly, the rattling of the sleeve **70** in the direction of the axis S inside the fastening bolt **50** can be avoided. Therefore, the communication holes (the retard port **75**, the advance port **76**) provided at the sleeve **70** can be positioned at predetermined positions with respect to the passage of the fastening bolt **50** (the retard passage **55** and the advance passage **56**).

(166) In addition, the opening/closing operation can be reliably performed by moving the spool **80** by the movement amount set in advance, and the controllability of the spool **80** is increased. Moreover, since the snap ring **110** is fit into the annular groove **59***b* of the fastening bolt **50**, the impact resistance in the direction of the axis S is increased, and the sleeve **70** can be reliably positioned and fixed to the predetermined position.

(167) Also, in the filter **60**, with the annular plate part **61** being disposed to abut against the annular receiving part **53** and the plate spring pieces **63** being disposed to abut against the bottom end **70***a*, the annular plate part **61** can be brought into close contact with the annular receiving part **53**. Accordingly, sealing performance can be ensured, so that a foreign matter in the hydraulic oil flowing in from the opening part **52** does not flow to the downstream side of the filter **60** from the gap.

(168) Moreover, the filter part **62** of the filter **60** is formed to include multiple micro pores **62***a* formed through an etching process, and the plate spring pieces **63** are formed to extend from the outer edge of the annular plate part **61** and be bent.

(169) In this way, it is possible to form the micro pores **62***a* with a desired pore diameter by suitably adjusting the etching process with respect to the blank material Bm in which a metal plate spring material is punched, and it is possible to form the plate spring pieces **63** simply by performing a bending process, and it is possible to easily manufacture the filter **60** as a whole.

(170) In addition, in the region of the bottom end **70***a*, the sleeve **70** includes the cutout passages **71***a* formed on the outer wall **71** to supply the fluid passing through the filter **60** to the inlet (the supply port **74**), and the abutting part abutting against the plate spring pieces **63** of the filter **60**. The cutout passages include the two cutout passages **71***a*, **71***a* as the first cutout passage and the second cutout passage formed as separated from each other about the axis S. The abutting parts include the

narrow abutting part **70***a***4** located between the first cutout passage (the cutout passage **71***a*) and the second cutout passage (the cutout passage **71***a*) and the wide abutting part **70***a***5** away from the cutout passages. The plate spring piece **63** is formed to include the first plate spring piece **63***a*, the second plate spring piece **63***b*, and the third plate spring piece **63***c* disposed at equal intervals in the circumferential direction of the annular plate part **61**.
(171) Accordingly, while sufficiently ensuring the passage of the flow of the hydraulic oil, the spring force can be generated with the structure being simplified, and the entire region of the annular plate part **61** in the circumferential direction can be in close contact with the annular receiving part **53**.
(172) In addition, the fluid control valve V includes the check valve **100** disposed on the inner side of the sleeve **70** to allow the hydraulic oil to flow in from the inlet (the supply port **74**). Therefore, compared with a configuration such as conventional examples in which the check valve is provided on the outer side of the sleeve **70**, the size can be reduced, and the integration level can be increased.
(173) In addition, the snap ring **110** has the discharge port (the notch part **110***a* and the circular holes **112***a*, **112***b*) able to discharge the hydraulic oil flowing through the discharge passage (the first communication path **71***d*, the second communication path **71***e*, and the communication concave part **71***h*) formed in the passage member (the fastening bolt **50**). Therefore, at the time of changing the valve timing in the valve timing change device M, extra hydraulic oil can be smoothly discharged without remaining, and a desired change operation can be attained.
(174) In addition, the snap ring **110** is formed in a plate shape that expands in a direction perpendicular to the axis S. Therefore, compared with a bottomed cylindrical shape, the thickness in the direction of the axis S can be reduced, and the size of the fluid control valve unit U can be reduced.
(175) In the following, an operation of the valve timing change device M in which the fluid control valve unit U can be mounted.
(176) In the state in which the internal combustion engine is stopped, as shown in FIG. **21**, the vane rotor is locked at the intermediate position with respect to the housing rotor **10** by using the lock mechanism **40**.
(177) In this way, the internal combustion engine can be smoothly started without rattling of the vane rotor **20**, etc. When the internal combustion engine is stopped, except for the amount that leaks from gaps, etc., the hydraulic oil is generally filled in the retard chamber RC through the opening of the first valve part **82** at the inactive position (the state in which communication between the first communication path **71***d* and the through path **71***d*.sub.1 and the retard port **75** is cut off) and a backflow prevention function of the check valve **100**.
(178) In addition, with the starting of the internal combustion engine, the hydraulic oil supplied through the gap passage Cp defined by the passages **1***b*, **1***c*, the opening part **52**, the cutout part **70***a***6**, and the cutout passage **71***a* opens the check valve **100** and flows into the fluid control valve V from the supply port **74**, and is supplied to the retard chamber RC from the retard port **75** or to the advance chamber AC from the advance port **76**.
(179) Then, when the hydraulic pressure of the hydraulic oil guided to the lock mechanism **40** through the through path **54** and the communication path **27** reaches a releasable hydraulic oil, the lock pin **41** is detached from the lock hole **11***d*, and the locking is canceled. In addition, after the internal combustion engine is started, the position of the spool **80** of the fluid control valve V is properly controlled via the driving shaft **7***a* of the electromagnetic actuator **7**, and phase control is performed so that the vane rotor **20** and the camshaft **1** are toward the retard side or the advance side or maintained at a predetermined angle.
(180) Firstly, the operation when the internal combustion engine is at a low-speed operation state is described. In the low-speed operation state, following the torque change (ΔT, −ΔT) applied by the camshaft, the hydraulic oil in the retard chamber RC and the advance chamber AC is able to be

reciprocated.
(181) For example, as shown in FIGS. **24** and **25**, in the case of the retard mode, the spool **80** is positioned at the inactive position due to the biasing force of the biasing spring **90**.
(182) In the retard mode, the first valve part **82** is set to the valve-open state in which the passage between the supply port **74** and the retard port **75** is opened, the second valve part **83** is set to the valve-closed state in which the passage between the supply port **74** and the advance port **76** is blocked. Specifically, the second land **83***a***1** of the second fixed part **83***a* opens the advance port **76** and the second cover part **83***b*.sub.2 of the second movable part **83***b* blocks the second inner passage **83***a*.sub.3. In addition, the discharge passage (the first communication path **71***d* and the through path **71***d*.sub.2) is in a state of being able to communicate with the advance port **76** to discharge the hydraulic oil in the advance chamber AC.
(183) In the state, when the camshaft **1** receives a reverse torque (−ΔT) with respect to the forward rotation direction CR, the hydraulic pressure of the hydraulic oil in the advance chamber AC increases. Therefore, as shown in FIG. **24**, the hydraulic oil inside the advance chamber AC causes the second cover part **83***b*.sub.2 of the second movable part **83***b* to be detached from the second fixed part **83***a* while resisting the biasing force of the compression spring **84**. Accordingly, the second inner passage **83***a*.sub.3 is opened and the hydraulic oil actively flows into the retard port **75** from the advance port **76**. At this time, the hydraulic oil in an amount less than the hydraulic oil toward the retard port **75** is discharged from the discharge port (the notch part **110***a*, the circular hole **112***a*) of the snap ring **110** through the discharge passage (the through path **71***d***2** and the first communication path **71***d* and the communication concave part **71***h*). In addition, the hydraulic oil leaked to the region where the biasing spring **90** is disposed is appropriately discharged from the discharge ports (the notch part **110***a*, the circular hole **112***b*) of the snap ring **110** through the discharge passage (the through path **71***e*.sub.1 and the second communication path **71***e*, and the communication concave part **71***h*).
(184) Meanwhile, when the camshaft **1** receives a torque (ΔT) in a forward direction, the hydraulic pressure of the hydraulic oil in the retard chamber RC increases. However, as shown in FIG. **25**, since the hydraulic oil in the retard chamber RC acts in a direction of bringing the second movable part **83***b* to abut against the second fixed part **83***a*, the second inner passage **83***a***3** is blocked, and the hydraulic oil from the retard port **75** toward the advance port **76** does not flow.
(185) By continuously receiving the reverse torque (−ΔT) and the forward torque (ΔT), the hydraulic oil in the advance chamber AC moves into the retard chamber RC, and the vane rotor is positioned at the most retard position as shown in FIG. **22**. In this process, in order to fill the hydraulic oil, the check valve **100** is opened appropriately and allows the hydraulic oil from the supply port **74** to flow in.
(186) Then, in the case of the advance mode, as shown in FIGS. **26** and **27**, the spool **80** resists the biasing force of the biasing spring **90** and is positioned to the innermost position in the direction of the axis S through the driving shaft **7***a* of the electromagnetic actuator **7**.
(187) In the advance mode, the second valve part **83** is set to the valve-open state in which the passage between the supply port **74** and the advance port **76** is opened, the first valve part **82** is set to the valve-closed state in which the path between the supply port **74** and the retard port **75** is blocked. Specifically, the first land **82***a*.sub.1 of the first fixed part **82***a* opens the retard port **75** and the first cover part **82***b*.sub.2 of the first movable part **82***b* blocks the first inner passage **82***a*.sub.3. In addition, the discharge passage (the communication path **71***d* and the through path **71***d*.sub.1) is in a state of being able to communicate with the retard port **75** to discharge the hydraulic oil in the retard chamber RC.
(188) In the state, when the camshaft **1** receives a reverse torque (−ΔT) with respect to the forward rotation direction CR, the hydraulic pressure of the hydraulic oil in the advance chamber AC increases. However, as shown in FIG. **26**, since the hydraulic oil in the advance chamber AC acts in a direction of bringing the first movable part **82***b* to abut against the first fixed part **82***a*, the first

inner passage **82***a*.sub.3 is blocked, and the hydraulic oil from the advance port **76** toward the retard port **75** does not flow.
(189) Meanwhile, when the camshaft **1** receives a torque (ΔT) in a forward direction, the hydraulic pressure of the hydraulic oil in the retard chamber RC increases. Therefore, as shown in FIG. **27**, the hydraulic oil inside the retard chamber RC causes the first cover part **82***b*.sub.2 of the first movable part **82***b* to be detached from the first fixed part **82***a* while resisting the biasing force of the compression spring **84**. Accordingly, the first inner passage **82***a***3** is opened and the hydraulic oil actively flows from the retard port **75** toward the advance port **76**. At this time, the hydraulic oil in an amount less than the hydraulic oil toward the advance port **76** is discharged from the discharge ports (the notch part **110***a*, the circular hole **112***a*) of the snap ring **110** through the discharge passage (the through path **71***d*.sub.1 and the first communication path **71***d* and the communication concave part **71***h*). In addition, the hydraulic oil leaked to the region where the biasing spring **90** is disposed is appropriately discharged from the discharge ports (the notch part **110***a*, the circular hole **112***b*) of the snap ring **110** through the discharge passage (the through path **71***e*.sub.1 and the second communication path **71***e*, and the communication concave part **71***h*).
(190) By continuously receiving the reverse torque (−ΔT) and the forward torque (ΔT), the hydraulic oil in the retard chamber RC moves into the advance chamber AC, and the vane rotor is positioned at the most advance position as shown in FIG. **23**. In this process, in order to fill the hydraulic oil, the check valve **100** is opened appropriately and allows the hydraulic oil from the supply port **74** to flow in.
(191) That is, the spool **80** of the fluid control valve V is formed so that, in the state of being positioned in the retard mode in which the first valve part **82** is opened and the second valve part **83** is closed, when the camshaft **1** receives the reverse torque (−ΔT), the second valve part **83** is opened to allow the flow of the hydraulic oil from the advance port **76** toward the retard port **75**, and in the state of being positioned at the advance mode in which the first valve part **82** is closed and the second valve part **83** is opened, when the camshaft **1** receives the forward torque (ΔT), the first valve part **82** is opened to allow the flow of the hydraulic oil from the retard port **75** toward the advance port **76**.
(192) Although the series of operations are operations when the internal combustion engine is in the low-speed operation state, for example, when the internal combustion engine is in a high-speed operation state, the torque change (ΔT, −ΔT) applied by the camshaft **1** decreases, no reciprocation of the hydraulic oil in the retard chamber RC and the advance chamber AC is generated, and it is difficult to open/close the first valve part **82** and the second valve part **83** through the torque change.
(193) As a result, through the opening of the check valve **100**, the hydraulic oil supplied from the supply port **74** actively flows into the retard chamber RC or the advance chamber AC, whereas the hydraulic oil in the advance chamber AC or the retard chamber RC is actively discharged outside from the discharge port (the notch part **110***a*, the circular hole **112***a*) of the snap ring **110** through the discharge passage (the through path **71***d*.sub.2 or the through path **71***d*.sub.1 and the first communication path **71***d*). In addition, the hydraulic oil leaked to the region where the biasing spring **90** is disposed is appropriately discharged from the discharge ports (the notch part **110***a*, the circular hole **112***b*) of the snap ring **110** through the discharge passage (the through path **71***e*.sub.1 and the second communication path **71***e*, and the communication concave part **71***h*).
(194) Then, in the case of the neutral hold mode, as shown in FIGS. **28** and **29**, the spool **80** resists the biasing force of the biasing spring **90** and is positioned to the intermediate position in the direction of the axis S through the driving shaft **7***a* of the electromagnetic actuator **7**.
(195) In the neutral hold mode, the first valve part **82** is set to the valve-closed state in which the passage between the supply port **74** and the retard port **75** is blocked, and the second valve part **83** is set to the valve-closed state in which the passage between the supply port **74** and the advance port **76** is blocked.

(196) Specifically, the first valve part **82** is set to a state in which the first land **82***a* of the first fixed part **82***a* blocks the retard port **75** and the first cover part **82***b*.sub.2 of the first movable part **82***b* blocks the first inner passage **82***a*.sub.3. In addition, the second valve part **83** is set to a state in which the second land **83***a*.sub.1 of the second fixed part **83***a* blocks the advance port **76** and the second cover part **83***b*.sub.2 of the second movable part **83***b* blocks the second inner passage **83***a*.sub.3. In addition, the communication between the discharge passage (the communication path **71***d* and the through path **71***d*.sub.1) and the retard port **75** is cut off, and the communication between the discharge passage (the communication path **71***d* and the through path **71***d*.sub.2) and the advance port **76** is cut off.
(197) In the state, when the camshaft **1** receives a reverse torque (−ΔT) with respect to the forward rotation direction CR, the hydraulic pressure of the hydraulic oil in the advance chamber AC increases. However, as shown in FIG. **28**, the advance port **76** is blocked by the second land **83***a*.sub.1 of the second valve part **83**. Therefore, the hydraulic oil in the advance chamber AC stays in the advance chamber AC without being able to move from the advance port **76** toward the retard port **75**.
(198) Meanwhile, when the camshaft **1** receives a torque (ΔT) in a forward direction, the hydraulic pressure of the hydraulic oil in the retard chamber RC increases. However, as shown in FIG. **29**, the retard port **75** is blocked by the first land **82***a***1** of the first valve part **82**. Therefore, the hydraulic oil in the retard chamber RC stays in the retard chamber RC without being able to move from the retard port **75** toward the advance port **76**.
(199) As described above, in the neutral hold mode, the reciprocation of the hydraulic oil between the retard chamber RC and the advance chamber AC is cut off, and the discharge passage (the through path **71***d*.sub.1, **71***d*.sub.2) is also blocked. Therefore, the vane rotor **20** is held at a desired intermediate position between the most retard position and the most advance position with respect to the housing rotor **10**.
(200) That is, in the fluid control valve V, in the state of being positioned in the neutral hold mode in which the first valve part **82** blocks the retard port **75** and the second valve part **83** blocks the advance port **76**, the spool **80** is formed to cut off the reciprocation of the hydraulic oil between the retard chamber RC and the advance chamber AC.
(201) As described above, the fluid control valve V is a torque-driven and hydraulic pressure-driven type fluid control valve able to reciprocate the hydraulic oil between the retard chamber RC and the advance chamber AC and discharge a portion of the hydraulic oil that is supplied by using a variable torque received by the camshaft **1**. Therefore, it is possible to change the opening/closing timing of the valve to a desired timing by reciprocating the hydraulic oil between the retard chamber RC and the advance chamber AC in an operation state (at the time of low-speed operation, etc.) in which a sufficient variable torque can be obtained and actively discharging the hydraulic oil in an operation state (e.g., at the time of high-speed operation) in which it is difficult to obtain a sufficient variable torque.
(202) FIG. **30** illustrates another embodiment of the filter included in the fluid control valve unit U. The components same as those of the filter are labeled with the same symbols, and the description thereof is omitted.
(203) As shown in FIG. **30**, a filter **160** according to the embodiment includes the annular plate part **61**, the filter part **62**, the plate spring pieces **63** (the first plate spring piece **63***a*, the second plate spring piece **63***b*, the third plate spring piece **63***c*), and a pair of clamping pieces **63***a*.sub.1 integrally formed with the first plate spring piece **63***a*.
(204) In addition, in the filter **160**, in the state of being assembled to the bottom end **70***a* of the sleeve **70**, the first plate spring piece **63***a* abuts against the narrow abutting part **70***a***4**, the second plate spring piece **63***b* and the third plate spring piece **63***c* abut against the wide abutting part **70***a*.sub.5. In the assembled state, the pair of clamping pieces **63***a*.sub.1 are assembled to sandwich the narrow abutting part **70***a*.sub.4 from the two sides. Therefore, the filter **160** can be positioned at

a predetermined angle about the axis S. Accordingly, the first plate spring piece **63***a* can be assembled to reliably abut against the narrow abutting part **70***a*.sub.4 without falling off from the narrow abutting part **70***a*.sub.4.
(205) In the embodiments, configurations in which the filter **60**, **160** includes three plate spring pieces (the first plate spring piece **63***a*, the second plate spring piece **63***b*, and the third plate spring piece **63***c*) as the plate spring pieces **63** are shown. However, the invention is not limited thereto. The plate spring pieces in other numbers may also be adopted.
(206) In the embodiment, configurations in which, in the filter **60**, **160**, the annular plate part **61** abuts against the annular receiving part **53** of the passage member (the fastening bolt **50**) and the plate spring pieces **63** abut against the bottom end **70***a* (the narrow abutting part **70***a*.sub.4, the wide abutting part **70***a*.sub.5) of the sleeve **70** are shown. However, if the sealing function can be ensured, on the contrary, a configuration in which the plate spring pieces **63** abut against the annular receiving part **53** of the passage member (the fastening bolt **50**) and the annular plate part **61** abuts against the bottom end **70***a* of the sleeve **70** (the narrow abutting part **70***a*.sub.4, the wide abutting part **70***a*.sub.5) may also be adopted.
(207) As an example, a configuration in which the portion serving for the seal function, such as the cylindrical part in the outer circumferential region of the annular plate part **61**, is integrally provided with the filter, and an inner circumferential wall in close contact with the cylindrical member is provided continuously with the annular receiving part can be adopted, for example.
(208) In the embodiments, the case where the lock mechanism **40** is locked at the intermediate position is shown. However, the invention is not limited thereto. The lock mechanism **40** may also be locked at the most retard position or other positions.
(209) In the embodiments, as a rotational biasing spring that rotationally biases the vane rotor **20**, the rotation biasing spring **30** applying a biasing force in the advance direction is shown. However, the invention is not limited thereto. On the contrary, a rotation biasing spring applying a biasing force in the opposite direction, i.e., the retard direction, may also be adopted.
(210) In the embodiments, the torque-driven and hydraulic pressure-driven type fluid control valve V is shown as the fluid control valve. However, the invention is not limited thereto. A fluid control valve of other embodiments may also be adopted as long as such fluid control valve supplies and discharges the hydraulic oil.
(211) In the embodiment, as the fluid control valve unit, the fluid control valve unit U in which the fluid control valve V is disposed on the inner side of the fastening bolt **50** as the passage member is shown. However, the invention is not limited thereto. The invention is applicable even to a configuration in which the fluid control valve V is disposed in another passage member or a cylinder block of an engine.
(212) In the embodiments, the case of handling hydraulic oil as a fluid controlled by the fluid control unit is shown. However, the invention is not limited thereto. The invention is also applicable at the time of controlling the flow of other fluids.
(213) As described above, the fluid control valve unit of the invention can simplify the structure, reduce the number of components, reduce the cost, and simplify the assembling process, and can prevent a foreign matter from entering, and hold the sleeve at a predetermined position to ensure functional reliability. Therefore, in addition to being applicable to an internal combustion engine mounted in the automobile, etc., the fluid control valve unit of the invention can also be used in an internal combustion engine mounted in a motorcycle, etc., or a machine or apparatus that controls the flow of other fluids.

Claims

1. A fluid control valve unit, comprising: a fluid control valve, comprising: a sleeve, having a bottomed cylindrical shape, having: an inlet; a communication port in communication with outside;

a bottom end; and an opening end, and defining an axis; and a spool slidably disposed in the sleeve to open and close the communication port; a passage member, having a cylindrical shape and provided with: an inner circumferential surface, fit with the sleeve; a passage of a fluid, leading to the communication port; an annular receiving part, facing the bottom end in a direction of the axis; and an opening part formed in adjacency with the annular receiving part and allowing the fluid to flow in; a filter, disposed between the bottom end and the annular receiving part; and a stopper, receiving the opening end of the sleeve accommodated in the passage member, wherein the filter is integrally formed by a metal plate spring member to include an annular plate part, a filter part surrounded by the annular plate part, and a plate spring piece extending from the annular plate part and applying a biasing force in the direction of the axis.

2. The fluid control valve unit as claimed in claim 1, wherein the passage member has an annular groove adjacent to the opening end in the direction of the axis and recessed with respect to the inner circumferential surface, and the stopper is a snap ring fit into the annular groove to detachably receive the spool.

3. The fluid control valve unit as claimed in claim 2, wherein the snap ring has a discharge port able to discharge the fluid flowing through a discharge passage formed in the passage member.

4. The fluid control valve unit as claimed in claim 3, wherein the snap ring is formed in a plate shape expanding in a direction perpendicular to the axis.

5. The fluid control valve unit as claimed in claim 2, wherein the sleeve comprises, as the communication port, a first communication port and a second communication port located on two sides sandwiching the inlet in the direction of the axis, and the spool comprises: a rod, reciprocating in the sleeve; a first valve part, provided at the rod to open and close a passage between the inlet and the first communication port; a second valve part, provided at the rod to open and close a passage between the inlet and the second communication port; and a biasing spring, applying a biasing force in a direction that brings the first valve part to abut against the snap ring.

6. The fluid control valve unit as claimed in claim 5, wherein the spool comprises a compression spring disposed between the first valve part and the second valve part, the first valve part comprises: a first fixed part, having a first land able to block the first communication port and a first inner passage formed on an inner side of the first land, and being fixed to the rod; and a first movable part, having a first cover part opening and closing the first inner passage and movably supported along the rod, the second valve part comprises: a second fixed part, having a second land able to block the second communication port and a second inner passage formed on an inner side of the second land, and being fixed to the rod; and a second movable part, having a second cover part opening and closing the second inner passage and movably supported along the rod, and the compression spring is disposed to apply a biasing force to close the first cover part and close the second cover part.

7. A valve timing change device, changing an opening/closing timing of an intake air valve or an exhaust air valve driven by a camshaft, the valve timing change device comprising: a housing rotor, rotating coaxially with the camshaft; a vane rotor, cooperating with the housing rotor to define an advance chamber and a retard chamber and integrally rotating with the camshaft; the fluid control valve unit as claimed in claim 5, configured to control supplying and discharging of hydraulic oil with respect to the advance chamber and the retard chamber, wherein the inlet of the fluid control valve unit is a supply port to which the hydraulic oil is supplied, the first communication port of the fluid control valve unit is a retard port in communication with the retard chamber, and the second communication port of the fluid control valve unit is an advance port in communication with the advance chamber.

8. The valve timing change device as claimed in claim 7, comprising a fastening bolt fastening the vane rotor to the camshaft, wherein the fastening bolt serves as the passage member of the fluid control valve unit.

9. The valve timing change device as claimed in claim 8, wherein the fluid control valve of the

fluid control valve unit is a torque-driven and hydraulic pressure-driven type fluid control valve reciprocating the hydraulic oil between the retard chamber and the advance chamber by using a variable torque received by the camshaft and being able to discharge a portion of the hydraulic oil that is supplied.

10. The valve timing change device as claimed in claim 9, wherein the spool is formed so that in a state of being positioned in a retard mode in which the first valve part is opened and the second valve part is closed, when the camshaft receives a reverse torque, the second valve part is opened to allow a flow of the hydraulic oil from the advance port toward the retard port, and in a state of being positioned in an advance mode in which the first valve part is closed and the second valve part is opened, when the camshaft receives a forward torque, the first valve part is opened to allow a flow of the hydraulic oil from the retard port toward the advance port.

11. The valve timing change device as claimed in claim 10, wherein the spool is formed so that in a state of being positioned in a neutral hold mode in which the first valve part blocks the retard port and the second valve part blocks the advance port, reciprocation of the hydraulic oil between the retard chamber and the advance chamber is cut off.

12. The fluid control valve unit as claimed in claim 1, wherein the filter is disposed so that the annular plate part abuts against the annular receiving part, and the plate spring piece abuts against the bottom end.

13. The fluid control valve unit as claimed in claim 12, wherein the sleeve comprises, in a region of the bottom end, a cutout passage formed on an outer wall to supply the fluid passing through the filter to the inlet; and an abutting part abutting against the plate spring piece of the filter.

14. The fluid control valve unit as claimed in claim 13, wherein the cutout passage comprises a first cutout passage and a second cutout passage formed as separated from each other about the axis, the abutting part comprises: a narrow abutting part located between the first cutout passage and the second cutout passage; and a wide abutting part away from the cutout passage, and the plate spring piece comprises a first plate spring piece, a second plate spring piece, and a third plate spring piece disposed at equal intervals in a circumferential direction of the annular plate part.

15. The fluid control valve unit as claimed in claim 14, wherein the first plate spring piece comprises a pair of clamping pieces abutting against the narrow abutting part and sandwiching the narrow abutting part.

16. The fluid control valve unit as claimed in claim 1, wherein the filter part comprises a plurality of micro pores formed through an etching process.

17. The fluid control valve unit as claimed in claim 1, wherein the plate spring piece is formed to extend from an outer edge of the annular plate part and be bent.

18. The fluid control valve unit as claimed in claim 1, wherein the fluid control valve comprises a check valve disposed on an inner side of the sleeve to allow the fluid to flow in from the inlet.